

Continuous optimization ENT305A

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Organization

Organization:

- Class 1: lecture + programming exercises
- Class 2: lecture (1h 30) + programming exercises (2h)
- Class 3: lecture (1h 30) + programming exercises (2h)
- Class 4: programming exercises
- Class 5: programming exercises (1h 30) + exam (2h)

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Course website:

<https://grosjean1.github.io/teaching/ensta/>

Course objectives and ENT306 project

Pre-requisite:

- Programming: little (python)
- Maths: little (Topology & Differential calculus).

Course objectives and ENT306 project

Skills to be developed:

- recognize when a practical question can be formulated as a **minimization problem**;
- formulate **constrained optimization problems** in Python and solve them as a black box with **AMPL**;
- solve **unconstrained problems** using classical optimization methods.

ENT306 project: two applications in energy

- energy distribution in a microgrid or macrogrid;
- fault detection for an energy-consuming device.

1 General introduction

- What is an optimization problem?
- Classes of problems
- Existence of a solution
- Derivatives

2 Methods for unconstrained optimization

- Optimality conditions
- Gradient methods
- Newton's method

3 Optimality conditions for constrained problems

- Linear constraints
- Non-linear constraints
- Sensitivity analysis

What is an optimization problem?

Definition 1

An **optimization problem** is a mathematical expression of the form:

$$\inf_{x \in \mathcal{D}} f(x), \quad \text{subject to: } x \in K, \quad (P)$$

where:

- $\mathcal{D}$ is a set called the **domain** of f ;
- $f: \mathcal{D} \rightarrow \mathbb{R}$ is called the **cost function** (or **objective function**);
- $K \subseteq \mathcal{D}$ is called **feasible set**.

In this class: $\mathcal{D} = \mathbb{R}^n$. **Unconstrained** optimization: $\mathcal{D} = K = \mathbb{R}^n$.

All results extend directly to **maximization** problems by replacing f with $-f$.

Abbreviation: “subject to” $\rightsquigarrow$ “s.t.”.

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Classes of optimization problems

From the point of view of **applications**, one can distinguish four classes of optimization problems.

- 1 Economic problems
- 2 Physical problems
- 3 Inverse problems
- 4 Learning problems.

1. Economic problems.

Any practical situation involving

- a **cost** to be minimized, or revenue or a performance index to be maximized;
- **operational decisions** (production levels in thermal power plants, water releases from a hydropower plant, maintenance schedules for a nuclear power plant, etc.);
- bounds on the decisions (which are often non-negative);
- **physical constraints** (“total production = demand”, “change in stock = inflow - outflow”, etc.).

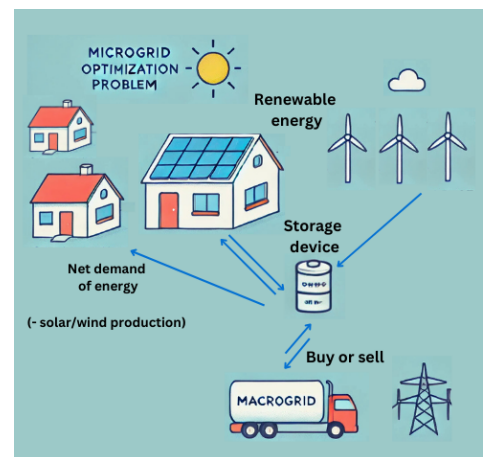
Optimization problem

Optimization variables:

- $x(s)$: state of charge of the battery.
- $a(s)$: amount of electricity *bought* on the network.
- $v(s)$: amount of energy *sold* on the network.

Parameters:

- $d(s)$: net demand of energy.
- $P_a(s)$: unit purchase price of energy at time s ;
- $P_v(s)$: unit selling price of energy at time s ;
- $x_{\max}$: storage capacity of the battery.



Horizon: 24 hours.

Time step: 1 hour.

Optimization over $T = 24$ intervals.

Optimization problem

Optimization variables:

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Parameters:

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- $x_{\max}$: storage capacity of the battery.

Constraints:

- $\forall s = 1, \dots, T,$

$$x(s+1) = x(s) - d(s) + a(s) - v(s)$$

- $x(1) = 0$
- $a(s) \geq 0, \forall s = 1, \dots, T$
- $v(s) \geq 0, \forall s = 1, \dots, T$
- $0 \leq x(s) \leq x_{\max}, \forall s = 1, \dots, T + 1.$

Cost function to be minimized:

$$J(x, a, v) = \sum_{s=1}^T (P_a(s)a(s) - P_v(s)v(s))$$

Classes of optimization problems

2. Physical problems.

Some equilibrium problems in **physics** can be formulated as optimization problems, involving an energy to be minimized.

- Mechanical structures
- Electricity networks
- Gas networks

Some similar problems arise in **economics** and game theory:

- Traffic models on road networks.

Classes of optimization problems

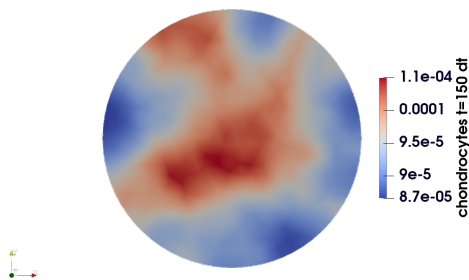
3. Inverse problems

Context. An unknown variable x must be identified from observations y related to x by $y = F(x)$.

Examples:

- parameter identification in tissue regeneration;
- estimation of the epicenter x of an earthquake from seismic measurements y ;
- localization of a crack x in a mechanical structure from displacement measurements y provided by sensors;
- estimation of the temperature in the core of a nuclear plant from external temperature measurements.

Example: an inverse problem in tissue regeneration



$$\partial_t c_1 = a_1 \Delta c_1 - \nabla \cdot (b c_1 \nabla h) - \alpha c_1 + \beta c_1 (1 - c_1 - c_2),$$

$$\partial_t c_2 = a_2 \Delta c_2 + \alpha c_1 - \delta c_2,$$

$$\partial_t h = -\gamma_1 h c_2 + \gamma_2 \frac{c_2}{(K_{c_1} + c_2)},$$

$$-\frac{\partial c_1}{\partial \nu} + b c_1 \frac{\partial h}{\partial \nu} = \frac{\partial c_2}{\partial \nu} = 0,$$

$$c_{10}(x, 0) = c_{10}(x), \quad c_{20}(x, 0) = c_{20}(x), \quad h(x, 0) = h_0(x),$$

Classes of optimization problems

The equation $y = F(x)$, with unknown x ,

- may not have a solution because measurements are inaccurate;
- may have several solutions (too few measurements).

This leads to the optimization problem

$$\inf_{x \in \mathcal{D}} \|y - F(x)\|^2, \quad \text{subject to: } x \in K,$$

where the constraints may model a priori knowledge on x .

Classes of optimization problems

4. Learning problems

Context. A heuristical relation must be found between two (or more) variables $x \in X$ and $y \in Y$, with the help of measurements $(x_k, y_k)_{k=1, \dots, K}$.

Examples.

- the size of a population in function of time
- the efficiency of solar pannel in function of its age, temperature, enlightment, etc
- the identity of a person on a picture.

Exercise

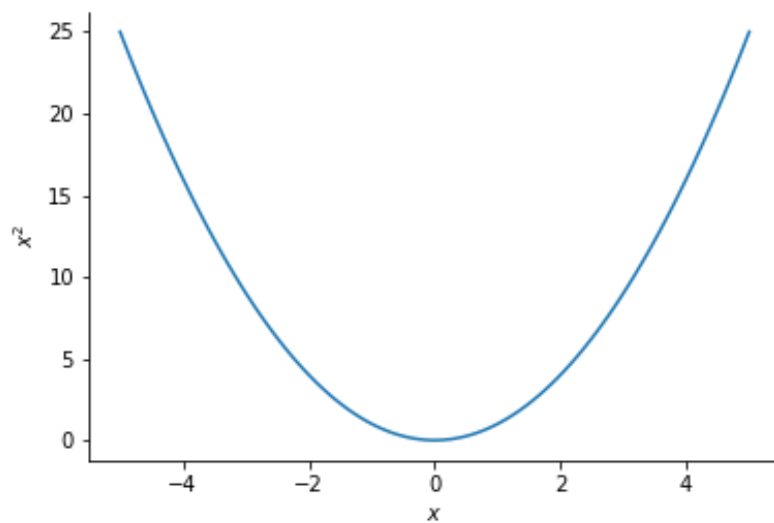
For each situation, decide whether it is an optimization problem.

- 1 Choose a thermostat setting between 19°C and 21°C to minimize energy consumption.
- 2 Estimate tomorrow's temperature from previous measurements.
- 3 Choose a route to arrive as early as possible while avoiding toll roads.

What is an optimization problem?

$$f : x \rightarrow x^2$$

$$\inf_{x \in \mathbb{R}} f(x), \text{ s.t. } x \in [-5, 5]$$



What is an optimization problem?

Definition 2

- A point x is called **feasible** if $x \in K$.
- A feasible point $\bar{x}$ is called **(global) solution** (to problem P) if

$$f(x) \geq f(\bar{x}), \quad \text{for all } x \in K.$$

- If $\bar{x}$ is a global solution, then $f(\bar{x})$ is the **value** of the optimization problem, denoted $\text{val}(P)$.

Example. The point $x = \pi$ is the solution of the problem

$$\inf_{x \in \mathbb{R}} \cos(x), \quad x \in [0, 2\pi].$$

What is an optimization problem?

Remarks.

- An optimization problem may **not** have a solution.

Examples:

$$\inf_{x \in \mathbb{R}} e^x, \quad (P_1)$$

$$\inf_{x \in \mathbb{R}} x^3. \quad (P_2)$$

- The **value** of an optimization problem is defined even when no solution exists; it belongs to $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$. In particular:

$$\text{val}(P_1) = 0, \quad \text{val}(P_2) = -\infty.$$

What is an optimization problem?

Definition 3

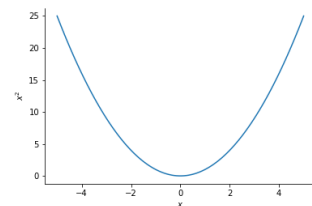
Let $\bar{x} \in K$. We call $\bar{x}$ a **local solution** to (P) if there exists $\varepsilon > 0$ such that, for all $x \in K$,

$$\|x - \bar{x}\| \leq \varepsilon \implies f(x) \geq f(\bar{x}).$$

Example:

$$\inf_{x \in \mathbb{R}} -x^2, \quad \text{s.t. } x \in [-1, 2].$$

The two local solutions are -1 and 2 ;
only 2 is a global solution.



What is an optimization problem?

Remarks.

- A global solution is also a local solution.
- In a finite-dimensional vector space, the notion of local optimality does not depend on the chosen norm.

What is an optimization problem?

Notation.

Let $\bar{B}(\bar{x}, \varepsilon)$ denote the closed ball of center $\bar{x}$ and radius ε .

Equivalent definition.

A feasible point $\bar{x}$ is a local solution to (P) if and only if there exists $\varepsilon > 0$ such that $\bar{x}$ is a **global** solution to the following **localized** problem:

$$\inf_{x \in \mathbb{R}^n} f(x), \quad x \in K \cap \bar{B}(\bar{x}, \varepsilon).$$

What is an optimization problem?

Constraints.

If we have constraints, they are equality or inequality constraints.

The feasible set K is described by

$$K = \left\{ x \in \mathbb{R}^n \mid \begin{array}{l} h_i(x) = 0, \quad \forall i \in \mathcal{E} \\ g_j(x) \leq 0, \quad \forall j \in \mathcal{I} \end{array} \right\},$$

where $h: \mathbb{R}^n \rightarrow \mathbb{R}^{m_1}$, $g: \mathbb{R}^n \rightarrow \mathbb{R}^{m_2}$.

We call the expressions

- $h_i(x) = 0$: **equality constraint**
- $g_j(x) \leq 0$: **inequality constraint.**

Course roadmap

	Necessary conditions	Sufficient conditions
Abstract formulation (exist.)		if ... then at least one solution
		if then at least one solution

	Necessary conditions	Sufficient conditions
No constraints $K = \mathbb{R}^d$ (opt.)	if then ...	if then ...
Affine constraints	...	... then global sol.
Non-linear constraints	...	... , ... then global sol.

Course roadmap

	Find a local solution
No constraints	methods
Affine constraints	
Non-linear constraints	

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Existence of a solution

Theorem 4 (existence of extreme value (Weierstrass))

Assume the following:

- K is **non-empty** and **compact** (i.e. closed and bounded)
- f is **continuous** on K .

Then the optimization problem (P) has (at least) one solution.

Remarks. If $K = \{x \in \mathbb{R}^n \mid h_i(x) = 0, \forall i \in \mathcal{E}, g_j(x) \leq 0, \forall j \in \mathcal{I}\}$, where h_i, g_j are continuous, then K is closed. In practical exercises, it is not necessary to justify the continuity of h_i or g_j .

Exercise

For each problem, decide whether Weierstrass' theorem applies and whether a minimizer exists.

A. $\min_{x \in [-2, 2]} x^2,$

B. $\inf_{x \in \mathbb{R}} e^x,$

C. $\min_{x \in [0, +\infty)} (x - 3)^2.$

Exercise

For each problem, decide whether Weierstrass' theorem applies and whether a minimizer exists.

A. $\min_{x \in [-2, 2]} x^2,$

B. $\inf_{x \in \mathbb{R}} e^x,$

C. $\min_{x \in [0, +\infty)} (x - 3)^2.$

Answer

- A: the theorem applies, and the minimizer is $x = 0$.
- B: the feasible set is not compact, and no minimizer exists ($\inf e^x = 0$).
- C: the theorem does not apply because the feasible set is unbounded, but the minimizer $x = 3$ exists.

The assumptions of Weierstrass' theorem are sufficient, not necessary.

Example

Exercise. Consider for $x \in \mathbb{R}^n$, the problem

$$\min \langle c, x \rangle, \quad \text{s.t. } \|x\|_2 \leq 1, \quad x \geq 0.$$

Prove that the problem is well posed and provide an obvious global solution if $c \geq 0$.

Example

Solution. The constraint set $K = \{x \in \mathbb{R}^n : x \geq 0, \|x\|_2 \leq 1\}$ is bounded because it is contained in the closed unit ball. It is also closed in $\mathbb{R}^n$ by continuity of the functions involved. Hence, it is compact, and since the objective function is continuous, the problem is well posed. In the case where $c \geq 0$, we have $\langle c, x \rangle \geq 0$ for all $x \in K$. At $x = 0 \in K$, the objective function equals 0. Therefore, 0 is a global minimizer.

Existence of a solution

Definition 5

We say that $f : \mathbb{R}^d \rightarrow \overline{\mathbb{R}}$ is **coercive** if the following holds: for any sequence $(x_k)_{k \in \mathbb{N}}$ in $\mathbb{R}^d$,

$$\|x_k\| \rightarrow \infty \implies f(x_k) \rightarrow +\infty.$$

Remark. The definition is independent of the norm.

Existence of a solution

Exercise. Consider

$$f: (x, y) \in \mathbb{R}^2 \mapsto x^4 - 2xy + 2y^2.$$

Prove that f is coercive on $\mathbb{R}^2$.

Existence of a solution

Solution. We have $x^4 \geq 2x^2 - 1$, since

$$0 \leq (x^2 - 1)^2 = x^4 - 2x^2 + 1.$$

Therefore

$$\begin{aligned} f(x, y) &\geq 2x^2 - 1 - 2xy + 2y^2 \\ &= (x^2 + y^2) - 1 + (x - y)^2 \\ &\geq \|(x, y)\|^2 - 1 \xrightarrow{\|(x, y)\| \rightarrow \infty} \infty, \end{aligned}$$

where $\|\cdot\|$ denotes the Euclidean norm. Thus f is coercive.

Existence of a solution

Lemma 6

Assume the following:

- K is **non-empty** and **closed**
- f is **continuous** on K
- f is **coercive** on K .

Then the optimization problem (P) has (at least) one solution.

Existence of a solution

Elements of proof.

Fix $x_0 \in K$. Since f is coercive, there exists $R > 0$ such that

$$x \in K, \|x - x_0\| \geq R \implies f(x) \geq f(x_0).$$

The set $K \cap \overline{B}(x_0, R)$ is non-empty and compact. By Weierstrass' theorem, f has a minimizer x^* on this set. Since x_0 belongs to the set, $f(x^*) \leq f(x_0)$. Hence $f(x^*) \leq f(x)$ both inside and outside the ball, so x^* is a global minimizer on K .

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Derivatives

Definition 7

A function $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **differentiable** at $\bar{x}$ if there exists a linear map $DF(\bar{x}): \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$F(\bar{x} + h) = F(\bar{x}) + DF(\bar{x})h + o(\|h\|).$$

The matrix

$$DF(\bar{x}) = \left(\frac{\partial F_i}{\partial x_j}(\bar{x}) \right)_{\substack{i=1,\dots,m \\ j=1,\dots,n}} \in \mathbb{R}^{m \times n}$$

is called **Jacobian** matrix.

Derivatives

- The function F is said to be continuously differentiable if the Jacobian $DF: x \in \mathbb{R}^n \rightarrow \mathbb{R}^{m \times n}$ is continuous.
- **Chain rule.** Let $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and let $G: \mathbb{R}^p \rightarrow \mathbb{R}^n$ be continuously differentiable functions. Let $H = F \circ G$ (that is, $H(x) = F(G(x))$). Then

$$DH(x) = DF(G(x))DG(x), \quad \text{for all } x \in \mathbb{R}^p.$$

Derivatives

Definition 8

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at $x \in \mathbb{R}^n$. We call **gradient** of f (at x) the column vector:

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x) \end{pmatrix} = Df(x)^\top.$$

Derivatives

Definition 9

The function $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **twice differentiable** if it is differentiable and DF is differentiable.

We denote: $\frac{\partial^2 F_i}{\partial x_j \partial x_k}(x) = \frac{\partial}{\partial x_j} \left(\frac{\partial F_i}{\partial x_k} \right)(x)$.

If $m = 1$, the matrix

$$D^2F(x) = \left(\frac{\partial^2 F}{\partial x_j \partial x_k}(x) \right)_{\substack{j=1,\dots,n \\ k=1,\dots,n}}$$

is called **Hessian** matrix. It is symmetric if F is twice continuously differentiable.

Derivatives

Exercise.

Calculate the gradient and the Hessian of the function

$$f: x \in \mathbb{R}^n \mapsto \frac{1}{2} \langle x, Ax \rangle + \langle b, x \rangle,$$

where $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$.

Derivatives

Solution. We have

$$f(x) = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j + \sum_{i=1}^n b_i x_i.$$

Therefore,

$$\begin{aligned} \frac{\partial f}{\partial x_k}(x) &= \frac{1}{2} \sum_{j=1}^n A_{kj} x_j + \frac{1}{2} \sum_{i=1}^n A_{ik} x_i + b_k \\ &= \frac{1}{2} (Ax)_k + \frac{1}{2} (A^\top x)_k + b_k. \end{aligned}$$

Therefore,

$$\nabla f(x) = \frac{1}{2} (A + A^\top) x + b.$$

Hessian: $D^2 f(x) = \frac{1}{2} (A + A^\top).$

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Optimality conditions

Let us fix a continuously differentiable function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ for the whole section. Let us consider

$$\inf_{x \in \mathbb{R}^n} f(x) \quad (P)$$

The function f is said to be **stationary** at $x \in \mathbb{R}^n$ if $\nabla f(x) = 0$.

Theorem 10 (Necessary optimality condition)

Let $\bar{x} \in \mathbb{R}^n$ be a **local solution** of (P) . Then, f is **stationary** at $\bar{x}$.

Remark. Stationarity is only a necessary condition!

Optimality conditions

Theorem 11

Assume that f is twice continuously differentiable. Let $\bar{x}$ be a stationary point.

■ **Necessary condition.**

If $\bar{x}$ is a **local solution** of (P) , then $D^2f(\bar{x})$ is **positive semi-definite**, that is to say,

$$\langle h, D^2f(\bar{x})h \rangle \geq 0, \quad \text{for all } h \in \mathbb{R}^n.$$

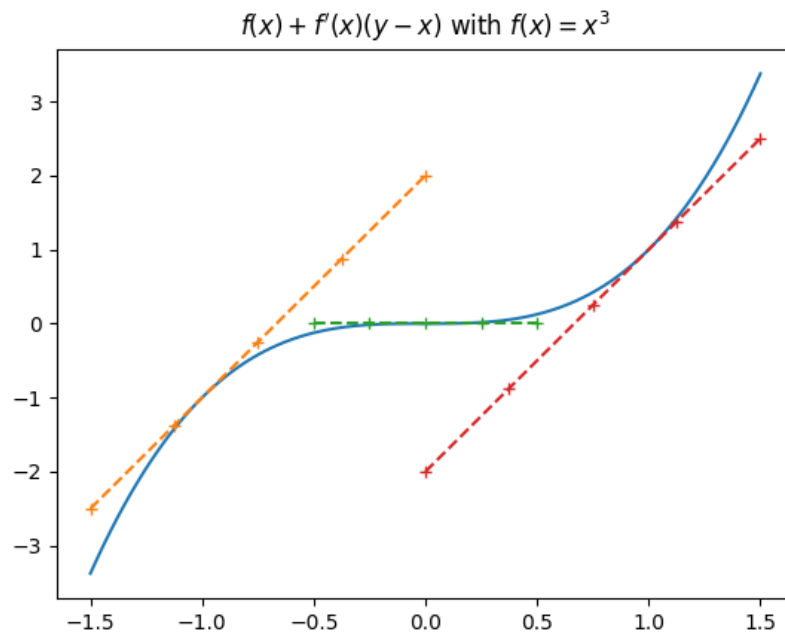
■ **Sufficient condition.**

If $D^2f(\bar{x})$ is **positive definite**, that is to say if

$$\langle h, D^2f(\bar{x})h \rangle > 0, \quad \text{for all } h \in \mathbb{R}^n \setminus \{0\},$$

then $\bar{x}$ is a **strict local solution** of (P) .

Illustration



Optimality conditions

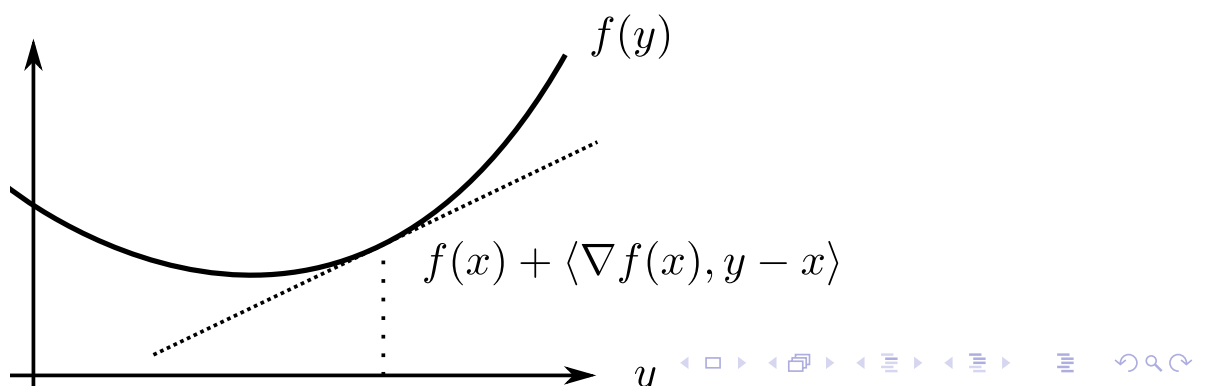
Theorem 12

- The function f is convex if and only if

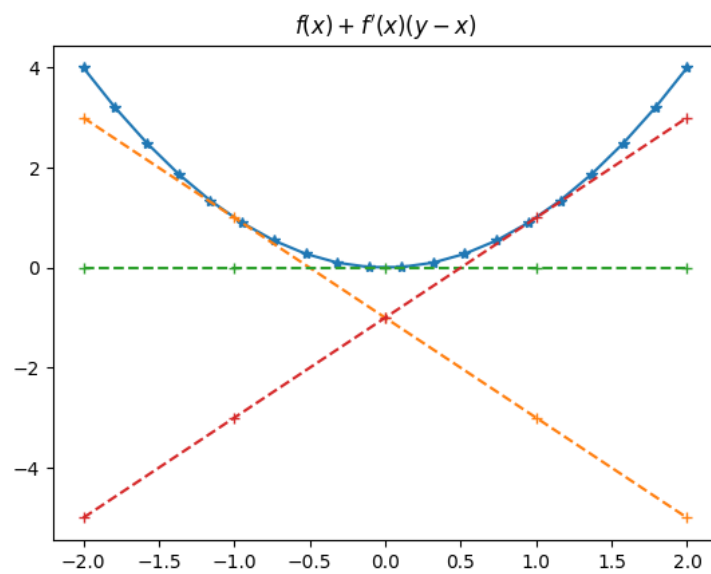
$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle,$$

for all x and $y \in \mathbb{R}^n$.

- If f is twice differentiable, then f is convex if and only if $D^2f(x)$ is symmetric positive semi-definite for all $x \in \mathbb{R}^n$.



Optimality conditions



Optimality conditions

Theorem 13

Assume that f is **convex**. Let $\bar{x}$ be a **stationary** point of f . Then it is a **global solution** of (P) .

Proof. For all $x \in \mathbb{R}^n$, we have

$$f(x) \geq f(\bar{x}) + \langle \nabla f(\bar{x}), x - \bar{x} \rangle = f(\bar{x}).$$

Optimality conditions

Exercise.

Let $A \in \mathbb{R}^{n \times n}$ be symmetric positive definite and let $b \in \mathbb{R}^n$. Let

$$f: x \in \mathbb{R}^n \mapsto \frac{1}{2} \langle x, Ax \rangle + \langle b, x \rangle.$$

Prove that

$$\inf_{x \in \mathbb{R}^n} f(x)$$

has a unique solution.

Optimality conditions

Solution.

- We have $\nabla f(x) = Ax + b$ and $\nabla^2 f(x) = A$. Since A is symmetric positive definite, thus symmetric positive semi-definite, the function f is convex.
- For a convex function, a point is a solution if and only if it is a stationary point. Thus it suffices to prove the existence and uniqueness of a stationary point.
- We have

$$\begin{aligned}
 x \text{ is stationary} &\iff \nabla f(x) = 0 \\
 &\iff Ax + b = 0 \\
 &\iff x = -A^{-1}b.
 \end{aligned}$$

Therefore there is a unique stationary point, which concludes the proof.

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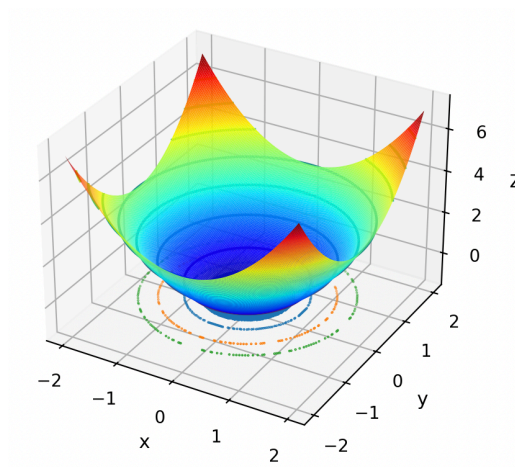
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Gradient methods



Our goal: solving **numerically** the problem

$$\inf_{x \in \mathbb{R}^n} f(x). \quad (P)$$

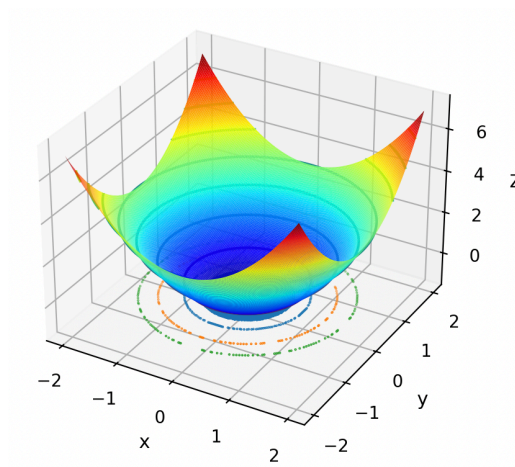
General idea: to compute a sequence $(x_k)_{k \in \mathbb{N}}$ such that

$$f(x_{k+1}) \leq f(x_k), \quad \forall k \in \mathbb{N},$$

the inequality being strict if $\nabla f(x_k) \neq 0$. $\rightarrow$ **Iterative** method.

How to compute x_{k+1} ?

Gradient methods



Our goal: solving **numerically** the problem

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General idea: to compute a sequence $(x_k)_{k \in \mathbb{N}}$ such that

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How to compute x_{k+1} ?

Gradient methods

Main idea of gradient methods.

Let $x_k \in \mathbb{R}^n$. Let d_k be a descent direction at x_k . Let $\alpha > 0$. Then

$$f(x_k + \alpha d_k) = f(x_k) + \alpha \underbrace{\langle \nabla f(x_k), d_k \rangle}_{<0} + o(\alpha).$$

Therefore, if α is small enough,

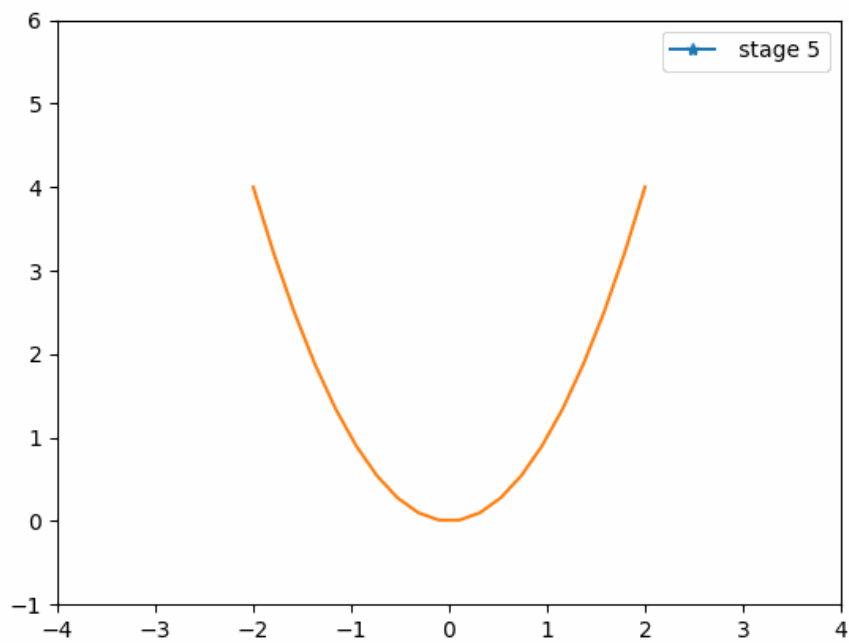
$$f(x_k + \alpha d_k) < f(x_k).$$

We can set

$$x_{k+1} = x_k + \alpha d_k.$$

Illustration

Gradient descent with $\alpha = 0.25$



Gradient methods

Definition 14

Let $x \in \mathbb{R}^n$ and let $d \in \mathbb{R}^n$. The vector d is called **descent direction** if

$$\langle \nabla f(x), d \rangle < 0.$$

Remark. If $\nabla f(x) \neq 0$, then $d = -\nabla f(x)$ is a descent direction. Indeed,

$$\langle \nabla f(x), d \rangle = -\langle \nabla f(x), \nabla f(x) \rangle = -\|\nabla f(x)\|^2 < 0.$$

Gradient methods

Gradient descent algorithm.

- 1 Input: $x_0 \in \mathbb{R}^n$, $\varepsilon > 0$.
- 2 Set $k = 0$.
- 3 While $\|\nabla f(x_k)\| \geq \varepsilon$, do
 - (a) Find a descent direction d_k .
 - (b) Find $\alpha_k > 0$ such that $f(x_k + \alpha_k d_k) < f(x_k)$.
 - (c) Set $x_{k+1} = x_k + \alpha_k d_k$.
 - (d) Set $k = k + 1$.
- 4 Output: x_k .

Remark. Step (b) is crucial; it is called **line search**.

The real number α_k is called the **step size**.

Exercise: Code the gradient descent algorithm.

Exercise

Consider

$$f(x) = (x - 3)^2, \quad x_0 = 0, \quad \alpha = \frac{1}{4},$$

and use

$$x_{k+1} = x_k - \alpha f'(x_k).$$

- 1 Should the first step go to the left or to the right?
- 2 Compute x_1 and x_2 .
- 3 Check that the cost decreases at each step.

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Answer

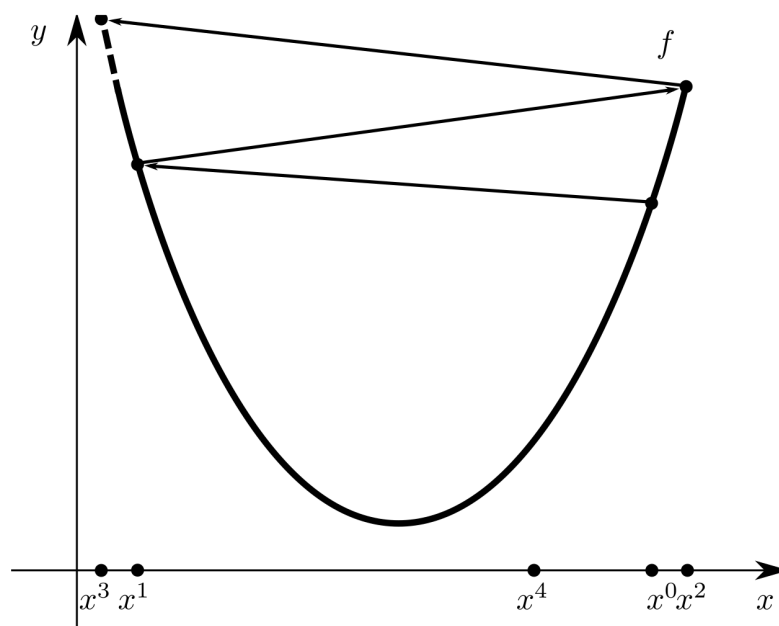
$f'(x) = 2(x - 3)$, so the first step goes to the right:

$$x_1 = \frac{3}{2}, \quad x_2 = \frac{9}{4}.$$

Moreover, $f(x_0) = 9$, $f(x_1) = \frac{9}{4}$ and $f(x_2) = \frac{9}{16}$.

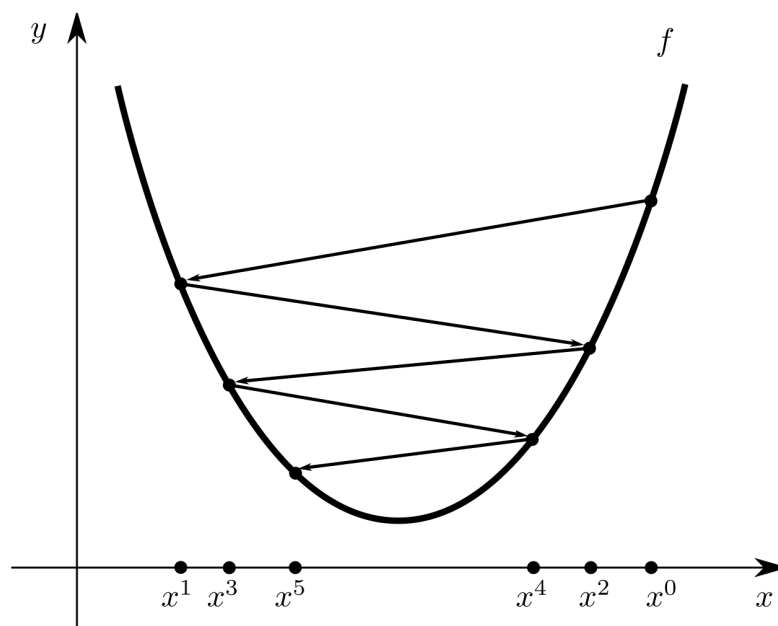
Gradient methods

On the choice of α_k .



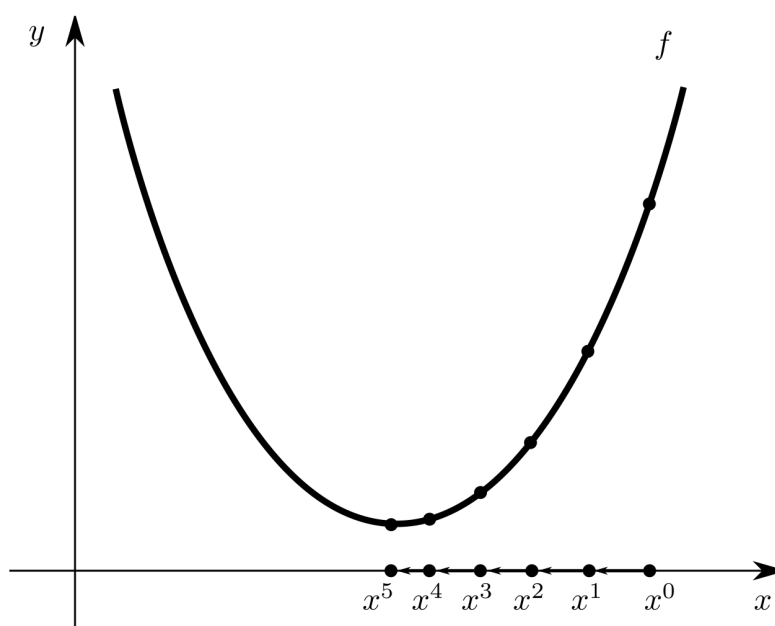
Gradient methods

On the choice of α_k .



Gradient methods

On the choice of α_k .



Gradient methods

On the choice of α_k .

Let us fix $x_k \in \mathbb{R}^n$. Let us define

$$\phi_k: \alpha \in \mathbb{R} \mapsto f(x_k + \alpha d_k).$$

The condition $f(x_k + \alpha_k d_k) < f(x_k)$ is equivalent to

$$\phi_k(\alpha_k) < \phi_k(0).$$

A natural idea: define α_k as a solution to

$$\inf_{\alpha \geq 0} \phi_k(\alpha).$$

Minimizing ϕ_k exactly may be too costly. A **compromise** is needed between computational effort and the quality of α .

Gradient methods

Observation. Recall that $\phi_k(\alpha) = f(x_k + \alpha d_k)$. We have

$$\phi'_k(\alpha) = \langle \nabla f(x_k + \alpha d_k), d_k \rangle.$$

In particular, since d_k is a descent direction,

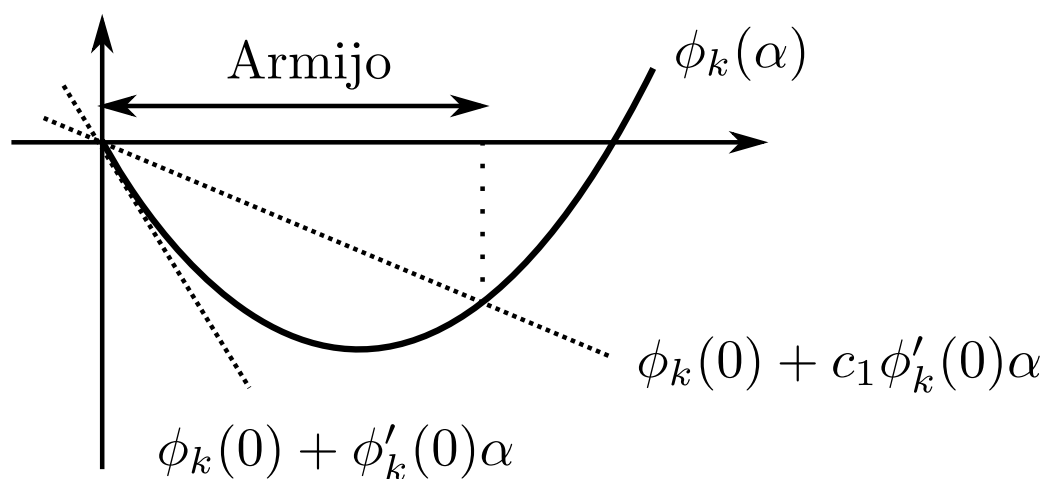
$$\phi'_k(0) = \langle \nabla f(x_k), d_k \rangle < 0.$$

Definition 15

Let us fix $0 < c_1 < 1$. We say that α satisfies **Armijo's rule** if

$$\phi_k(\alpha) \leq \phi_k(0) + c_1 \phi'_k(0) \alpha.$$

Gradient methods



Gradient methods

Backtracking algorithm for Armijo's rule

- 1 Input: $c_1 \in (0, 1)$, $\beta > 0$, and $\gamma \in (0, 1)$.
- 2 Set $\alpha = \beta$.
- 3 While α does not satisfy Armijo's rule,
 - Set $\alpha = \gamma\alpha$.
- 4 Output α .

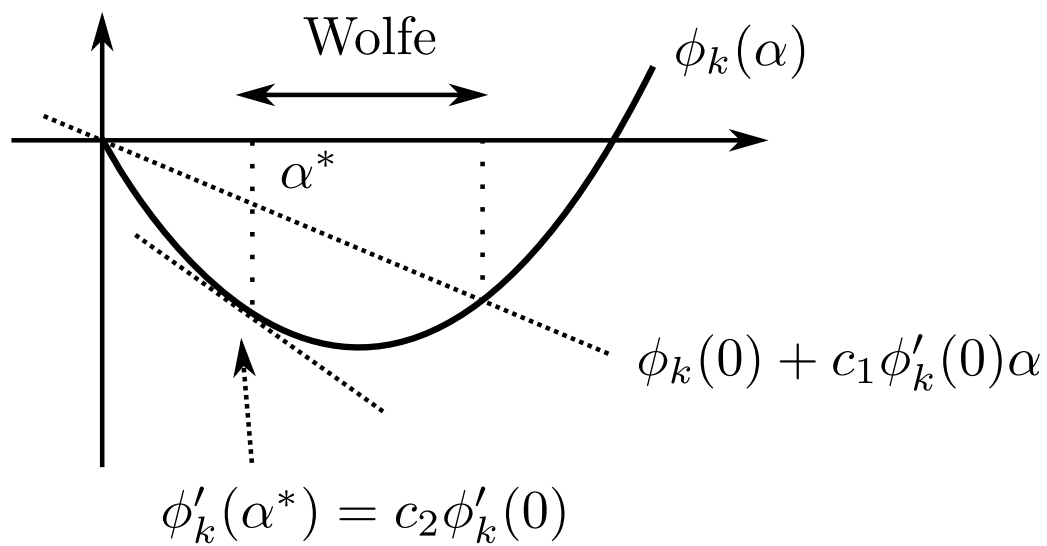
Gradient methods

Definition 16

Let $0 < c_1 < c_2 < 1$. We say that $\alpha > 0$ satisfies **Wolfe's rule** if

$$\phi_k(\alpha) \leq \phi_k(0) + c_1 \phi'_k(0) \alpha \quad \text{and} \quad \phi'_k(\alpha) \geq c_2 \phi'_k(0).$$

Gradient methods



Gradient methods

Bisection method for Wolfe's rule

- 1 Input: $c_1 \in (0, 1)$, $c_2 \in (c_1, 1)$, $\beta > 0$, $\alpha_{min}, \alpha_{max}$.
- 2 Set $\alpha = \beta$.

While Wolfe's rule is not satisfied:

- 1 if α does not satisfy Armijo's rule :
 - Set $\alpha_{max} = \alpha$
 - $\alpha = 0.5(\alpha_{min} + \alpha_{max})$
- 2 if α satisfies Armijo's rule and $\phi'_k(\alpha) < c_2\phi'_k(0)$:
 - Set $\alpha_{min} = \alpha$
 - $\alpha = 0.5(\alpha_{min} + \alpha_{max})$
- 3 Output: α .

Gradient methods

General comments on theoretical results from literature.

- Under standard assumptions, line-search algorithms find step sizes satisfying Armijo's or Wolfe's conditions in **finitely many iterations**.
- Without convexity assumption on f , very little can be said about the convergence of the sequence $(x_k)_{k \in \mathbb{N}}$. Typical results ensure that any accumulation point is stationary.
- In practice: $(x_k)_{k \in \mathbb{N}}$ “usually” **converges to a local solution**. Thus a good **initialization** (that is the choice of x_0) is crucial.
- In general, **slow** convergence.

1 General introduction

- What is an optimization problem?
- Classes of problems
- Existence of a solution
- Derivatives

2 Methods for unconstrained optimization

- Optimality conditions
- Gradient methods
- **Newton's method**

3 Optimality conditions for constrained problems

- Linear constraints
- Non-linear constraints
- Sensitivity analysis